Ben McKeon

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Research Interests

Numerical analysis: Finite-difference schemes, geometric numerical integration
Scientific computing: Splitting techniques, Poisson/symplectic integrators
Computational physics: Hamiltonian simulations, particle-in-cell codes

Education

Aug 2023 – Aug 2027 | **PhD Numerical Analysis** | **University of Limerick, Ireland**
Thesis: Geometric integration of Hamiltonian systems with multiple invariants
Supervisor: Dr Alan F. Hegarty
Funding: Research Ireland (grant no. SFI 18/CRT/6049)

Sep 2021 – Aug 2023 | **MSc Experimental Physics** | **University of Galway, Ireland**
Thesis: Gradient-index lenses: Symplectic ray tracing and optical testing
Supervisor: Dr Alexander V. Goncharov
Funding: University of Galway (College of Science and Engineering Scholarship)

Aug 2017 – Aug 2021 | **BSc Mathematics and Physics** | **University of Limerick, Ireland**
Thesis: Numerical analysis of symplectic integrators with applications in astrodynamics
Supervisor: Dr Alan F. Hegarty
Final grade: First Class Honours

Research Experience

Jan 2026 – Mar 2026 | **Rutherford Appleton Laboratory** | **Didcot, UK**
Investigated how the absence of splitting errors in Pauli algebra methods compares with the numerical errors present in standard integrators used for relativistic charged-particle dynamics.
Supervisor: Dr Holger Schmitz

Sep 2021 – Dec 2022 | **CSIRO Parkes Observatory** | **NSW, Australia (online)**
Coordinated with colleagues in Australia and the USA to conduct remote observations of nearby stars and the galactic centre on behalf of the Breakthrough Listen initiative.
Supervisor: Dr Howard Isaacson

Jun 2021 – Aug 2021 | **I-LOFAR Radio Telescope** | **Birr, Ireland**
Developed Python tools to detect anomalous signals in data collected by LOFAR stations in Birr (Ireland) and Onsala (Sweden), paving the way for future dual-site astrophysical surveys.
Supervisor: Dr Evan Keane

Publications

- **McKeon, B.** and Hegarty, A.F. Energy- and volume-preserving integrators for Hamiltonian systems with two or more degrees of freedom. In preparation for SIAM Journal on Numerical Analysis
- **McKeon, B.**, Hegarty, A.F. and Schmitz, H. Direct comparisons of covariant and lab-frame relativistic particle pushers at second and fourth order. In preparation for Journal of Scientific Computing
- **McKeon, B.** and Goncharov, A.V., 2023. Symplectic numerical methods in optics and imaging: ray tracing in spherical gradient-index lenses and computer-generated image rendering. Applied Optics, 62(32), pp. 8621–8631
- **McKeon, B.** and Goncharov, A.V., 2023. Performance Considerations for Ray Tracing in Gradient-index Optics with Symplectic Numerical Methods. In London Imaging Meeting (Vol. 4, pp. 35–39). Society for Imaging Science and Technology
- Johnson, O.A., Gajjar, V., Keane, E.F., McKenna, D.J., Giese, C., **McKeon, B.**, Carozzi, T.D., Alcaria, C., Brennan, A., Brzycki, B. and Croft, S., 2023. A Simultaneous Dual-site Technosignature Search Using International LOFAR Stations. The Astronomical Journal, 166(5), p. 193

Conference Presentations

Jul 2026	**Foundations of Computational Mathematics 2026** *A Large Step Size Integrator for Relativistic Charged-Particle Dynamics*	**Universität Wien, Austria**
May 2026	**22nd Workshop on Numerical Methods for Problems with Layers** *Higher-Order Methods as an Alternative to Adaptive Time Step Sub-Cycling*	**FernUniversität in Hagen, Germany**
Dec 2025	**Christmas High Power Laser Science Community Meeting** *Comparing and Evaluating a Selection of Implicit Particle Pushers*	**Abingdon School, UK**
Jun 2025	**30th Biennial Numerical Analysis Conference** *Pauli Matrices: A Better Basis for Relativistic Electrodynamics?*	**University of Strathclyde, UK**
Aug 2024	**Penn State SETI Center Symposium** *Comparing Numerical Ray Tracing Methods in the Presence of Strong Gravitational Lensing*	**Green Bank Observatory, USA**
Jun 2023	**London Imaging Meeting** *Performance Considerations for Ray Tracing in Gradient-index Optics with Symplectic Numerical Methods*	**Institute of Physics, UK**
Sep 2021	**Irish National Astronomy Meeting** *Are We Alone? A LOFAR SETI Survey of TESS-Selected Exoplanets*	**Online**
Mar 2021	**All-Ireland Conference of Undergraduate Research** *Orbit Modelling of the International Space Station Using Symplectic Integrators in Python*	**University of Limerick, Ireland**

Software

- **evolute** (**E**nergy- and **VOLU**me-preserving **T**ime integration **E**ngine): First-, second- and fourth-order variants of a novel non-symplectic scheme that conserves energy and phase-space volume. Benchmarked against symplectic and discrete-gradient methods.
- **particle-pushers**: Library for tracking relativistic charged particles in electromagnetic fields. Compares the behaviour of multiple lab-frame and covariant particle pushers at second and fourth order, complete with a seven-case benchmark suite.

Teaching Experience

2023–26	**Mathematics Tutor** Linear Algebra 1 Science Mathematics 2 Engineering Mathematics 3	**University of Limerick, Ireland**
2021–23	**Physics Laboratory Demonstrator** Introduction to Physics Experimental & Computational Physics	**University of Galway, Ireland**

Awards

PhysiGIST Graduate Research Communication Competition 2021: 3rd prize
Irish National Astronomy Meeting 2021: Peter Curran Award (shared with Owen Johnson)

Professional Service

Peer review: *Optics Letters*, *JOSA A*, *Applied Optics*, *Optics Continuum*
Conference organisation: IMS & Irish SIAM Student Chapter Conference 2025, University of Limerick

Public Engagement & Outreach

School visits: Talks for primary- and secondary-school students on SETI, maths content in UL courses and careers in maths and stats (Irish Astronomy Week 2024, Maths Week 2024, Science Week 2025, 2026)
Public lectures: Python Ireland 2023; Galway Astrofest 2024; UL Open Science Seminar 2024; Limerick Astronomy Club 2025